A Short Primer on Graph Neural Networks

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1. Introduction

Graph neural networks (GNNs) learn representations of nodes, edges and whole graphs by repeatedly exchanging information between neighbouring nodes. This process is called message passing. In each layer, every node aggregates messages from its neighbours, combines them with its own state, and applies a learned transformation.

2. Graph Convolutional Networks

A graph convolutional network (GCN) averages neighbour features using a symmetric normalisation based on node degrees, followed by a linear layer and a non-linearity. Because every neighbour is weighted by a fixed coefficient, GCN layers are cheap to compute and work well on homophilous graphs where connected nodes share labels.

Attention and Sampling

3. Graph Attention Networks
A graph attention network (GAT) replaces the fixed coefficients of a GCN with learned attention weights. Each node scores its neighbours with a small attention function and normalises the scores with a softmax. Multiple attention heads are usually concatenated or averaged. Attention lets the model down-weight noisy or irrelevant neighbours.

4. Neighbourhood Sampling
GraphSAGE-style models sample a fixed number of neighbours per node instead of using the full neighbourhood. Sampling bounds memory use on very large graphs and enables inductive learning on nodes that were not seen during training.

Limitations and Evaluation

5. Known Limitations of Message Passing
Over-smoothing: as more layers are stacked, node representations become increasingly similar, which hurts classification accuracy in deep GNNs.
Over-squashing: information from distant nodes must pass through narrow bottlenecks, so long-range dependencies are compressed into fixed-size vectors.
Expressiveness: standard message passing is at most as powerful as the 1-WL graph isomorphism test, so some non-isomorphic graphs cannot be distinguished.

6. Evaluation Practice
Results on small citation benchmarks vary noticeably with the random seed. Reports should therefore give the mean and standard deviation over several seeds, state whether lower or higher values of each metric are better, and use the same data splits for every model.